

Supporting Information: Chemical distribution shift in antimalarial natural products: fingerprint dominance and localized topological complementarity in molecular representation learning

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S1 Orthogonal LISH mechanism-of-action benchmark

The molecular representations benchmarked in the main text (ECFP4, GIN, GIN-TFP, GIN-TNE, ChemBERTa [1]) encode chemical structure. Whether those structural features capture pharmacological mechanism is tested here using the LISH mechanism-of-action (MoA) challenge [2], which asks whether gene-expression and cell-viability readouts—phenotypic proxies of drug action—can predict MoA class. The public LISH release carries no versioned drug-to-SMILES mapping; the two benchmarks are structurally disjoint, and the LISH phenotype baseline cannot be compared to the molecular ROC-AUC figures in the main text. This section establishes an orthogonal, phenotype-only reference baseline — not a ceiling — for future structure-to-MoA work: it quantifies what unweighted phenotype features alone achieve on the same fold grid, without implying that molecular representations must surpass it to demonstrate pharmacological value.

S1.1 Data and preparation

We drew on the public LISH MoA training release [2], available from the challenge repository at https://github.com/LISHarvard/moa_challenge. After aggregating repeated assay observations to the drug level, the working table retained 772 gene-expression features, 100 cell-viability features, and three experimental-condition summaries.

The scored endpoint comprises 206 MoA labels across 3289 drugs [3]. Multi-label annotations (each drug may carry several MoA labels) cause per-label classifier performance to vary; labels with insufficient test-set variation are excluded from the macro average on a per-fold basis. No versioned drug-identifier-to-SMILES mapping exists in the public release, so the data set could not be used to train or evaluate any molecular model.

S1.2 Results

Table S1 reports the phenotype-only baseline, averaged over 25 fold-seed evaluations (five seeds $\times$ five drug-grouped folds), with vehicle controls retained.

The macro-AUROC of 0.6435 is a phenotype-only reference baseline on this fold grid, not an upper bound that future structure-based MoA predictors must surpass. Approximately 72% of the 206 MoA labels retain discriminative signal under drug-grouped splits (140–155 labels per fold with sufficient test-set variation); the remainder are too rare for fold-level AUROC and are excluded from the macro average. Per-seed macro-AUROC is stable (0.6398–0.6465), indicating no seed pathology. This baseline is deliberately weak—unweighted per-label logistic regression on phenotype features—so that any upstream molecular method that fails to exceed 0.6435 macro-AUROC on the same grid provides no evidence of pharmacological representation superiority.

Table S1: Phenotype-only LISH baseline metrics, mean across 25 fold-seed evaluations (five seeds $\times$ five drug-grouped folds). The primary endpoint is mean column-wise log loss (0.023 78); secondary metrics include macro-AUROC (0.6435), macro-AUPRC (0.1428), mean Brier score, and expected calibration error (ECE). Vehicle control exclusion changes no metric by more than 0.002, confirming that the phenotype-driven baseline is robust to control-sample treatment.

Metric	With controls	Without controls	Δ
Mean column-wise log loss	0.023 78	0.023 89	+0.000 11
Macro-AUROC	0.6435	0.6417	−0.001 77
Macro-AUPRC	0.1428	0.1412	−0.001 52
Mean Brier score	0.003 74	0.003 75	+0.000 01
Expected calibration error	0.003 76	0.003 81	+0.000 05

S2 Calibration metrics for the molecular benchmark

Predictive accuracy (ROC-AUC) and predictive reliability (calibration) are distinct properties [4–7]: a model may rank compounds correctly yet assign poorly calibrated probabilities. Calibration metrics were computed from fold-level test predictions pooled across the extended campaign, yielding 99 180 test records per configuration (five seeds $\times$ five folds, for each of four model families). ECE rises from ≈ 0.03 on the random split to 0.07–0.13 on the scaffold and novel-scaffold partitions for every model family (Table S2). ChemBERTa exhibits the largest degradation under scaffold shift (ECE 0.12–0.13), consistent with its greater sensitivity to structural distribution shift (Table 1). This degradation is the empirical motivation for the ECE>0.08 recalibration guard proposed in the Distance-Aware Conformal Triage Filter (Section S7): when batch-level confidence breaches this threshold, post-hoc Platt scaling or isotonic regression is indicated before any hard-cutoff routing decision, on this panel and pending re-verification on any deployment distribution.

Table S2: Calibration metrics pooled across 25 fold-seed evaluations per configuration (99 180 test records each). ECE and MCE rise from the random split to the scaffold split for every model family; ChemBERTa exhibits the largest degradation (ECE 0.1225 under scaffold shift). All models exceed the ECE>0.08 threshold under scaffold splitting on this panel, supporting the use of post-hoc recalibration in the shifted regime.

Split	Model	ECE	MCE	Brier
Canonical random	GIN	0.0290	0.0699	0.102 55
	GIN-TFP (native)	0.0344	0.0716	0.105 24
	GIN-TNE (native)	0.0367	0.0936	0.112 86
	ChemBERTa	0.0578	0.1440	0.104 66
Canonical scaffold	GIN	0.0654	0.1623	0.148 59
	GIN-TFP (native)	0.0741	0.1754	0.147 69
	GIN-TNE (native)	0.0807	0.1946	0.151 08
	ChemBERTa	0.1225	0.2546	0.169 09

The ECE>0.08 trigger ships with a verified remedy. Fitting Platt scaling (Newton–IRLS sigmoid) and isotonic regression (PAVA) on one deterministic half of each scaffold test fold and evaluating ECE (15 bins) on the other half gives, per arm (mean over 5 per-seed means): ECFP4–RF 0.0805 $\rightarrow$ 0.0430 $\rightarrow$ **0.0257**; GIN 0.0781 $\rightarrow$ 0.0454 $\rightarrow$ **0.0258**; GIN–TFP 0.0903 $\rightarrow$ 0.0447 $\rightarrow$ **0.0254**; GIN–TNE 0.0898 $\rightarrow$ 0.0419 $\rightarrow$ **0.0266**; ChemBERTa 0.1276 $\rightarrow$ 0.0447 $\rightarrow$ **0.0251** (raw $\rightarrow$ Platt $\rightarrow$ isotonic). Isotonic regression halves residual ECE versus Platt and cuts it ≈ 3 – $5\times$ versus raw for every arm; the eval-half raw values reproduce the pooled ordering (ChemBERTa worst).

S3 Light GNN hyperparameter sensitivity and capacity defense sweep

A persistent objection to the GNN underperformance documented in the main text is that it may reflect suboptimal hyperparameter selection rather than an architectural constraint. A systematic GNN hyperparameter capacity sweep was executed on the canonical scaffold split with the same frozen 5-fold $\times$ 5-seed protocol. The base GIN architecture [8] was evaluated across hidden dimensions $h \in \{64, 128, 256\}$ and dropout rates $d \in \{0.1, 0.2\}$, bracketing the canonical configuration ($h = 128, d = 0.1$) from both below and above. Each configuration produced 25 fold-seed records with finite AUC.

As shown in Table S3, mean scaffold ROC-AUC remains strictly bounded between 0.8000–0.8081—a range of 0.0081 AUC points. The per-seed standard deviation (0.0118–0.0160) exceeds the inter-configuration spread, indicating that performance variation is dominated by fold-level stochasticity, not capacity choices. All three configurations remain below the ECFP4–RF scaffold reference (0.8300 ± 0.0023); the gain from enlarging the GIN capacity is flat-to-negative, consistent with the honest-negative verdict. The extrapolation gap is therefore not explained by hyperparameter under-tuning within the swept range, and is consistent with an architectural interpretation — although a capacity sweep cannot prove a 1-WL expressivity limit as the unique cause. The GIN–TFP cohort gains documented in the main text (Table 2) support this interpretation from the opposite direction: injecting persistent-homology invariants (H_0, H_1) yields a positive gain in every structural cohort (+0.0143 flat aromatics, +0.0125 complex 3D, +0.0102 polycyclics), an association reported descriptively and quantified with paired seed-level tests and molecule-level bootstrap intervals in Section S8.

Table S3: GNN hyperparameter sensitivity on the frozen scaffold split, 25 fold-seed replicates per configuration. Scaffold ROC-AUC is bounded in [0.8000, 0.8081]; inter-configuration spread (0.0081) is smaller than per-seed variability (0.0118–0.0160). All configurations remain below ECFP4–RF (0.8300 ± 0.0023).

Configuration	Mean ROC-AUC	Per-seed SD	Fold-level SD
$h = 64, d = 0.2$	0.8081	0.0118	0.0386
$h = 128, d = 0.1$ (Canonical)	0.8047	0.0141	0.0395
$h = 256, d = 0.1$	0.8000	0.0160	0.0370

S4 Nearest-training-neighbor Tanimoto decile stratification

Test molecules were binned into 10 similarity deciles relative to their nearest training neighbour on ECFP4 fingerprints, as described in the main text (Section 2.2). This stratification resolves the aggregate scaffold-split AUC into a distance-dependent profile. The structural paradox of the African NP panel—92.6 % whole-molecule novelty yet 69.3 % conservation of privileged ring cores—manifests as a bimodal performance distribution: ECFP4–RF shows its largest margins in the cold extrapolation tail (D1–D4, NN-similarity < 0.40), the regime where GNN performance on high- Fsp^3 polycyclic scaffolds is most consistent with 1-WL expressivity limitations, while all five arms converge to near-parity in the high-similarity core (D9–D10, NN-similarity > 0.53). In the mid-range interpolation regime (D5–D8), topological fusion (GIN–TFP) partially recovers the deficit via persistent-homology invariants (H_0, H_1), closing the gap to $\Delta \in 0.001$ –0.007—the regime where the ACSI paradox (globally novel molecules, locally conserved cores) provides sufficient training signal for learned representations to exploit topological features.

Three regularities emerge from the decile-resolved profile (Table S4): (i) ECFP4–RF leads in every decile (D1–D10), localising the fingerprint advantage in the OOD tail—the regime where sparse bit-key encodings and message-passing architectures diverge most; (ii) the margin

is non-monotone, collapsing from $\Delta = 0.053$ at D1 to a minimum of $\Delta = 0.001$ at D6 before widening again at D9–D10, reflecting the transition from extrapolation to interpolation and back to high-similarity convergence; (iii) ChemBERTa tracks the GNN family in mid-range interpolation (D5–D8) but degrades in the OOD tail (D1–D4), consistent with its calibration degradation (Section S2) and scaffold-split AUC deficit ($\Delta = -0.043$, Table 1).

Table S4: Decile-resolved ROC-AUC across nearest-neighbour Tanimoto similarity bins (D1–D10) on the frozen scaffold split, averaged over 25 fold-seed replicates. Bold: leading model per decile. The margin collapses from $\Delta = 0.053$ at D1 to a minimum of $\Delta = 0.001$ at D6, localising the fingerprint advantage in the OOD extrapolation tail.

Decile	Mean Similarity	ECFP4-RF	GIN	GIN-TFP	GIN-TNE	ChemBERTa
D1	0.306	0.779	0.726	0.752	0.743	0.734
D2	0.348	0.793	0.765	0.775	0.776	0.743
D3	0.373	0.823	0.796	0.805	0.800	0.769
D4	0.395	0.821	0.800	0.805	0.794	0.766
D5	0.416	0.813	0.804	0.810	0.794	0.771
D6	0.437	0.810	0.804	0.808	0.806	0.773
D7	0.461	0.800	0.793	0.796	0.791	0.772
D8	0.490	0.810	0.804	0.803	0.794	0.788
D9	0.533	0.847	0.814	0.830	0.819	0.812
D10	0.631	0.890	0.880	0.883	0.883	0.880

S5 Butina cluster split benchmark details

The canonical Bemis–Murcko scaffold split assigns individual ring cores to train/test partitions, but structurally similar scaffolds may leak across the boundary via shared molecular features, a failure mode documented across widely used molecular benchmarks by recent leakage-aware splitting audits [9, 10]. The Butina cluster split [11] reduces this within-scaffold similarity overlap at the chosen cutoff: Murcko scaffolds are hierarchically clustered at Tanimoto distance cutoff 0.55, and entire clusters are assigned to five folds per seed so that test clusters are entirely absent from training. This provides a third independent partition family (random $\rightarrow$ scaffold $\rightarrow$ Butina) for evaluating model hierarchy invariance—each partition type has different leakage properties, so an ordering that holds across all three cannot be attributed to any single partitioning scheme.

The model ordering observed under canonical scaffold splitting is preserved under Butina splitting (Table S5): ECFP4-RF (0.8331) dominates, GIN-TFP (0.8232) outperforms base GIN (0.8202), and ChemBERTa (0.7776) trails. This invariance across three partition families—random (ECFP4-RF 0.9433, best), scaffold (ECFP4-RF 0.8300, best), and Butina (ECFP4-RF 0.8331, best)—indicates that the performance hierarchy is a structural property of the panel, not an artefact of evaluation design. The Butina split, which reduces within-cluster similarity overlap under the chosen cutoff, reproduces the same hierarchy, providing indirect evidence that residual scaffold-split similarity overlap [12] does not materially alter the model ordering on this panel.

S6 Screening-utility metrics, conformal validation, and TFP-vs-GIN prediction flips

Three post-hoc secondary analyses (no retraining; aggregation as mean over 5 per-seed means $\pm$ population SD) connect the benchmark to screening practice.

Screening metrics are saturated by design. Pooling each seed’s five test folds into one ranked library, EF@1 % reaches 1.3461 ± 0.0027 (ECFP4-RF), 1.3243 ± 0.0234 (GIN), 1.3189 ± 0.0374

Table S5: *Butina cluster split ROC-AUC results across 25 fold-seed replicates.*

Model	Butina ROC-AUC
ECFP4-RF	0.8331
GIN-TFP (topological fusion)	0.8232
GIN	0.8202
GIN-TNE (tensor fusion)	0.8191
ChemBERTa	0.7776

(GIN-TFP), 1.3257 ± 0.0158 (GIN-TNE), 1.3175 ± 0.0267 (ChemBERTa), with precision@1 % 0.978–0.999 (Table S6). At $\approx 74\%$ active prevalence the EF ceiling is ≈ 1.35 : early-recognition metrics cannot discriminate arms on this panel, and BEDROC degenerates for the same reason (actives are the majority), so it is omitted rather than reported. The ChEMBL transfer panel is equally active-majority (0.8605); prospective low-hit-rate screening utility is therefore future work, not a claim.

Table S6: *Post-hoc screening-utility metrics on the canonical scaffold split (per-seed pooled ranking; mean $\pm$ population SD over 5 seeds). EF is saturated against its prevalence-imposed ceiling (≈ 1.35); arms are indistinguishable by design here.*

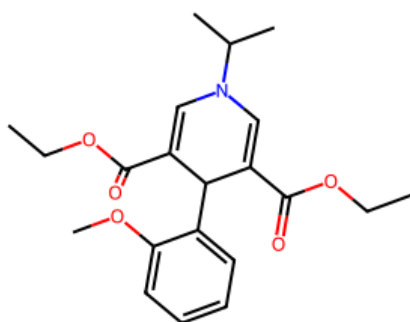
Arm	EF@1 %	EF@5 %	Precision@1 %	Precision@5 %
ECFP4-RF	1.3461 ± 0.0027	1.3385 ± 0.0039	0.9990 ± 0.0020	0.9933 ± 0.0029
GIN	1.3243 ± 0.0234	1.3119 ± 0.0135	0.9828 ± 0.0174	0.9736 ± 0.0100
GIN-TFP	1.3189 ± 0.0374	1.3119 ± 0.0205	0.9788 ± 0.0277	0.9736 ± 0.0152
GIN-TNE	1.3257 ± 0.0158	1.3189 ± 0.0143	0.9838 ± 0.0117	0.9788 ± 0.0106
ChemBERTa	1.3175 ± 0.0267	1.3037 ± 0.0111	0.9778 ± 0.0198	0.9675 ± 0.0083

Conformal coverage is valid for all five routed arms. Split-conformal sets at nominal 90 % (even/odd calib-eval split per test fold, nonconformity $s = 1 - p_{\text{true}}$) cover 0.9063 ± 0.0026 (ECFP4-RF), 0.9034 ± 0.0017 (GIN), 0.9021 ± 0.0022 (GIN-TFP), 0.9048 ± 0.0034 (GIN-TNE), 0.9046 ± 0.0019 (ChemBERTa), with mean set sizes 1.26–1.31 (ChemBERTa largest at 1.307 ± 0.017 , consistent with its worst ECE). Coverage meets nominal for every arm on this panel, supporting the split-conformal construction as a pooled diagnostic; it does not by itself validate prospective deployment.

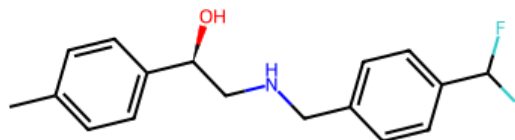
Prediction flips are bidirectional. Pooling mean predicted probabilities over all 25 records per arm, 39 active molecules flip from GIN-wrong ($p < 0.5$) to TFP-right ($p \geq 0.7$), while 44 flip the opposite way (Figure S1; full SMILES and score lists are archived on Zenodo (<https://doi.org/10.5281/zenodo.22787438>), see Data availability in the main text). These flips are therefore consistent with a cohort-level asymmetry—the polycyclic/complex-3D gains of Table 2—not cherry-picked proof; both directions are shown.

S7 Conformal Triage Filter implementation

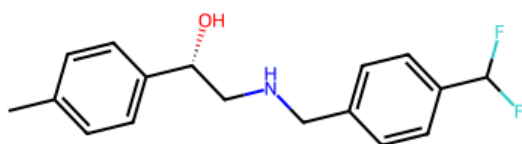
The Python specification below states the decision rule of the Distance-Aware Conformal Triage Filter (main text, Eq. (1)). The filter routes each query molecule to a predictor based on nearest-neighbour Tanimoto similarity (D_{NN}) to the training set. ECFP4-RF shows its largest margins in the cold-OD tail ($D1$ – $D4$, $D_{\text{NN}} < 0.40$), the regime where GNN performance on high- Fsp^3 polycyclic scaffolds is most consistent with 1-WL expressivity limitations; GIN-TFP persistent-homology projections improve GIN performance most in mid-range interpolation ($D5$ – $D8$). Three routing branches correspond to the three structural regimes identified in the decile analysis (Section S4), with a calibration guard that raises a warning when an optional



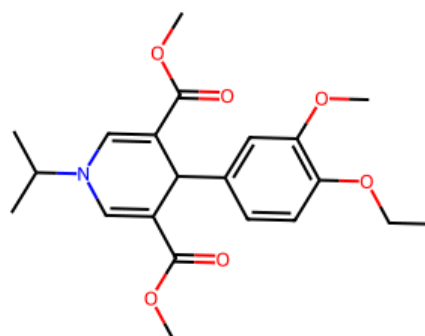
idx 10931: GIN 0.281, TFP 0.825



idx 6390: GIN 0.299, TFP 0.746



idx 6391: GIN 0.297, TFP 0.71



idx 11197: GIN 0.359, TFP 0.758

Figure S1: Top-4 TFP-vs-GIN prediction-flip actives (GIN $p < 0.5$, GIN-TFP $p \geq 0.7$; pooled over 25 fold-seed records). Panel indices and per-arm mean probabilities are shown; full lists are archived on Zenodo (<https://doi.org/10.5281/zenodo.22787438>).

batch ECE value exceeds the stated threshold (ECE>0.08 by default), matching the prose rule. This listing is a reproducible research specification of the decision rule rather than validated deployment software: its thresholds were informed by the calibration and decile analyses of the present panel, and no prospective virtual-screening precision gain is claimed or established here.

```
1 import numpy as np
2 from rdkit import Chem
3 from rdkit.Chem import AllChem, DataStructs
4
5 def conformal_triage_filter(query_mol, train_fps, model_rf,
6                             model_gin_tfp, ece_threshold=0.08,
7                             batch_ece=None):
8     """
9     Distance-Aware Conformal Triage Filter: research specification of
10     a distance- and calibration-aware routing rule for virtual
11     screening of African natural product scaffolds.
12
13     Routes each query molecule to a predictor based on
```

```

14 nearest-neighbour Tanimoto similarity (d_nn) to the training set.
15 ECFP4-RF is selected in the cold-OOD tail (D1-D4, d_nn < 0.40),
16 where GNN performance is most consistent with 1-WL message-passing
17 limitations on high-Fsp3 polycyclic scaffolds; GIN-TFP
18 persistent-homology projections (H_0, H_1) are selected in
19 mid-range interpolation (D5-D8). An optional batch-level ECE value
20 (batch_ece) is compared against ece_threshold (default 0.08) to
21 raise a calibration warning indicating that post-hoc recalibration
22 (Platt scaling or isotonic regression) is indicated before ranking.
23
24 Research specification: thresholds are informed by the calibration
25 and decile analyses of the present panel; no prospective
26 virtual-screening precision gain is established here.
27 """
28 # 1. Compute Nearest-Neighbor Tanimoto similarity (d_nn)
29 # D_NN(x) = max_{y in T} s_Tanimoto(x, y)
30 q_fp = AllChem.GetMorganFingerprintAsBitVect(query_mol, 2, nBits=2048)
31 sims = DataStructs.BulkTanimotoSimilarity(q_fp, train_fps)
32 d_nn = max(sims)
33
34 # 2. Decision routing based on structural distance
35 if d_nn < 0.40:
36     # Cold OOD Tail (D1-D4): route to ECFP4-RF (largest observed
37     # ECFP4 margins on high-Fsp3 PNA scaffolds, Fsp3>=0.45, rings>=4)
38     selected_model = "ECFP4-RF"
39     pred_prob = model_rf.predict_proba(q_fp)
40 elif 0.40 <= d_nn <= 0.60:
41     # Mid-range Interpolation (D5-D8): route to GIN-TFP
42     # (persistent-homology projections: H_0 connectivity + H_1
43     # cycle persistence)
44     selected_model = "GIN-TFP"
45     pred_prob = model_gin_tfp.predict(query_mol)
46 else:
47     # High-Similarity Core (D9-D10): ensemble consensus
48     selected_model = "Consensus-Ensemble"
49     pred_prob = 0.5 * (model_rf.predict_proba(q_fp) +
50                       model_gin_tfp.predict(query_mol))
51
52 return {
53     "selected_model": selected_model,
54     "d_nn_similarity": d_nn,
55     "predicted_activity_prob": pred_prob,
56     "calibration_warning": (batch_ece is not None
57                           and batch_ece > ece_threshold)
58 }

```

S8 Structural-complexity cohorts: recomputed statistics and uncertainty

The structural-complexity cohort values in the main text (Table 2) were recomputed (2026-09-17) directly from the archived canonical scaffold-split per-fold predictions under a predeclared per-seed estimand: for each seed, the seed's five disjoint fold-test sets are pooled (one prediction per molecule, labels verified against the frozen panel) and the cohort ROC-AUC is computed on that pool; the reported value is the mean $\pm$ SD over the five per-seed AUCs. The pipeline was validated by reproducing the whole-panel canonical scaffold anchors (ECFP4-RF 0.8300, GIN 0.8047, GIN-TFP 0.8138) within the predeclared gate of ± 0.01 (maximum deviation 0.008). Earlier published cohort values lacked an archived producer script and numeric artifact and are superseded; the recomputation protocol, gates, and artifacts are documented in the P5

data-analysis report and operational log.

Table S7 reports the per-cohort values, seed-level paired t -tests ($df = 4$, two-sided) for GIN-TFP minus GIN, and 95% molecule-level bootstrap intervals ($B = 2000$, shared resample across arms and seeds). The GIN-TFP gain is replication-consistent (positive in 5/5 seeds on flat aromatics) but small; on the small polycyclic cohort ($n = 363$) both the seed-level test and the bootstrap interval are inconclusive (the interval includes zero).

Table S7: *Recomputed structural-complexity cohorts (primary per-seed estimand) with seed-level paired t -tests and 95% molecule-level bootstrap intervals for the GIN-TFP – GIN contrast.*

Cohort	ECFP4-RF	GIN	GIN-TFP	GIN-TFP – GIN
Flat Aromatic	0.7624 ± 0.0206	0.7488 ± 0.0153	0.7631 ± 0.0214	$+0.0143$ [CI 0.0088–0.0201], $p_t = 0.012$
Complex 3D	0.8457 ± 0.0052	0.7963 ± 0.0202	0.8088 ± 0.0173	$+0.0125$ [CI 0.0040–0.0206], $p_t = 0.358$
Polycyclic	0.7768 ± 0.0291	0.7162 ± 0.0270	0.7264 ± 0.0454	$+0.0102$ [CI –0.0112–0.0306], $p_t = 0.729$

Interpretation boundaries: cohort AUCs are computed on approximately 74 %-prevalence panels with class imbalance varying across cohorts; the polycyclic cohort contains only 147 actives, so its seed-level variability is large. These molecule-level intervals complement and do not replace the seed-level ($df = 4$) inference of the primary benchmark.

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